

Acoustic Doppler Motion Estimation

Preliminary implementation and simulation results

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GitHub Project Site

<https://qxie123.github.io/elec5305-project-520096747/>

1. Objective and background

This project investigates motion estimation from the acoustic Doppler trajectory of a passing source recorded by a stationary microphone. Following Feedback 1, the focus has expanded from simulating and visualising Doppler shifts to estimating source speed, closest-approach distance and pass-by time. The longer-term research question is whether combining harmonic tracks improves robustness to noise and STFT resolution limitations.

The present checkpoint implements a single-tone, constant-speed, straight-line baseline. As the source approaches, its observed frequency exceeds the emitted frequency; after it passes, the observed frequency is lower. Frequency-ridge tracking turns this transition into measurements for a physical motion estimator.

2. Work completed to date

The preliminary MATLAB script, **run_doppler_demo.m**, has been added to the project repository with **README_PRELIMINARY.md** run instructions. The supplied MATLAB output figures and estimation-summary CSV demonstrate a completed noiseless baseline: received-signal synthesis, windowed FFT/STFT analysis, spectral peak interpolation and nonlinear parameter fitting. Noise robustness, multi-harmonic comparison and real-recording validation remain future work.

3. Methodology and implementation

Propagation model. Emission time u and reception time t are distinguished. For speed v , closest distance d and closest-approach emission time u_0 :

$$r(u) = \sqrt{d^2 + v^2(u - u_0)^2}$$

$$t = u + r(u)/c$$

$$f_{\text{received}}(t) = f_0 / [1 + r'(u)/c]$$

The subsonic retarded-time relation is inverted analytically. The received waveform uses phase $2\pi f_0 u$ and simplified relative amplitude $d/r(u)$. The closest-approach emission reaches the receiver at $u_0 + d/c$; this differs from the emission-time parameter.

Tracking and fitting. Each Hann-windowed frame is analysed using MATLAB `fft`. The dominant peak within $0.8f_0$ to $1.2f_0$ is refined by three-point log-magnitude interpolation. MATLAB `fminsearch` fits speed, distance and u_0 with multiple initial guesses. Positive parameters use log coordinates and a penalty restricts physically admissible values. The emitted frequency is assumed known.

The implementation was prepared with AI assistance. Standard MATLAB functions are used for FFT, optimisation and output; the student must review and understand the implementation and document subsequent changes.

4. Experimental settings and quantitative results

Parameter	Baseline setting
Sampling rate / duration	44,100 Hz / 10 s
Emitted frequency / sound speed	3,000 Hz / 343 m/s
Speed / closest distance / emission-time closest approach	10 m/s / 5 m / 5 s
Noise	No added noise (Inf dB SNR)
Hann window / hop / FFT length	4,096 / 256 / 16,384 samples

Zero padding supports finer spectral interpolation; the window duration determines the underlying time-frequency resolution. The numerical values below come from the supplied MATLAB estimation_summary.csv.

Parameter	True	Estimated	Absolute error
Speed (m/s)	10	10.000408935	0.000408935
Closest distance (m)	5	5.006280816	0.006280816
Closest emission time (s)	5	4.999988700	0.000011300

Frequency metric	Value (Hz)
Tracking RMSE against theoretical frequency	0.014747046
Tracking bias	-0.000127117
RMSE between extracted and fitted tracks	0.005947992

The relative speed error is 0.00409%, and the relative closest-distance error is 0.12562%. The fitted emission-time closest approach differs from the true setting by approximately 11.30 microseconds. These errors describe one ideal simulated run, not a statistical uncertainty estimate.

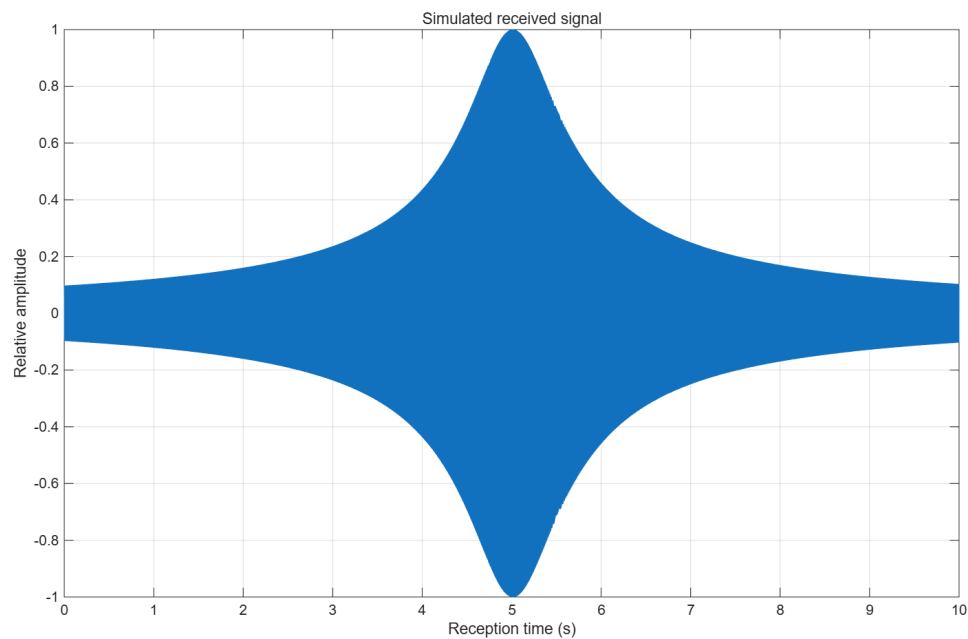


Figure 1. Simulated received waveform. Relative amplitude peaks near the closest pass-by because the model uses inverse-distance attenuation.

5. Time-frequency results and discussion

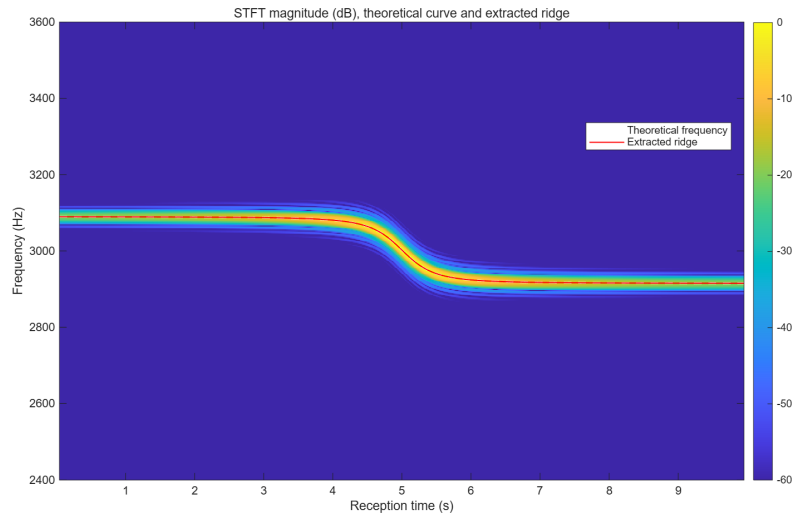


Figure 2. MATLAB STFT magnitude with the theoretical trajectory and extracted ridge. The frequency falls smoothly from approximately 3,090 Hz during approach to 2,915 Hz during recession.

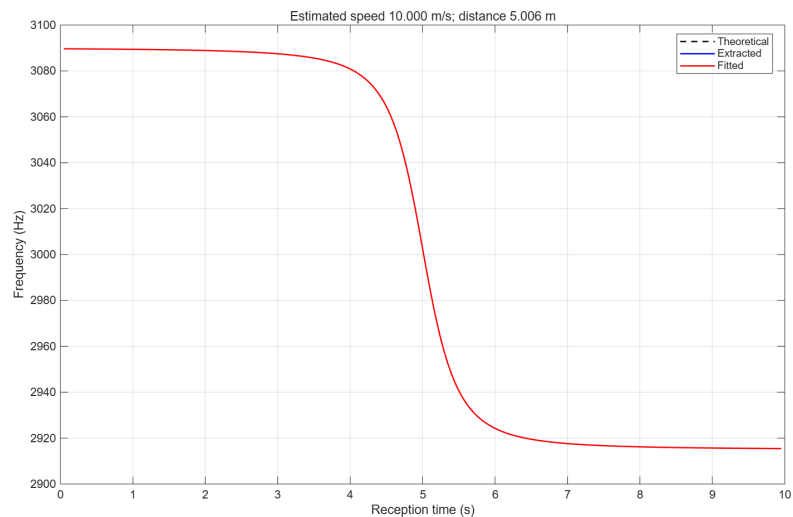


Figure 3. Theoretical, extracted and fitted frequency tracks nearly overlap at this plotting scale. The title reports rounded estimates; the table on page 2 retains greater numerical precision.

The low tracking RMSE and small parameter errors show that the baseline can recover the known motion under noiseless, matched-model conditions. The simulator and estimator share the same physical model, so this is a controlled consistency check. It does not establish robustness to reflections, source-frequency drift or unknown real-world geometry.

6. Next steps and conclusion

Next, compare STFT windows of 2,048, 4,096 and 8,192 samples, then test 20, 10, 5 and 0 dB SNR across repeated noise realisations. Evaluate tracking and parameter errors separately. Add multiple harmonics and compare their joint estimates against the single-tone baseline. Finally, attempt a controlled recording with independently measured source speed.

This checkpoint demonstrates an initial simulation-to-estimation pipeline with actual MATLAB output. The results provide a baseline for the planned resolution and noise-robustness experiments; multi-harmonic improvement remains a hypothesis to be tested.

Sources: supplied Craig Jin Project Feedback 1; supplied ELEC5305 Project Proposal Guide; supplied Feedback 2 instructions; student-supplied MATLAB figures and estimation_summary.csv (9 October 2026). Technical documentation: MathWorks fft and fminsearch documentation, linked in README_PRELIMINARY.md.